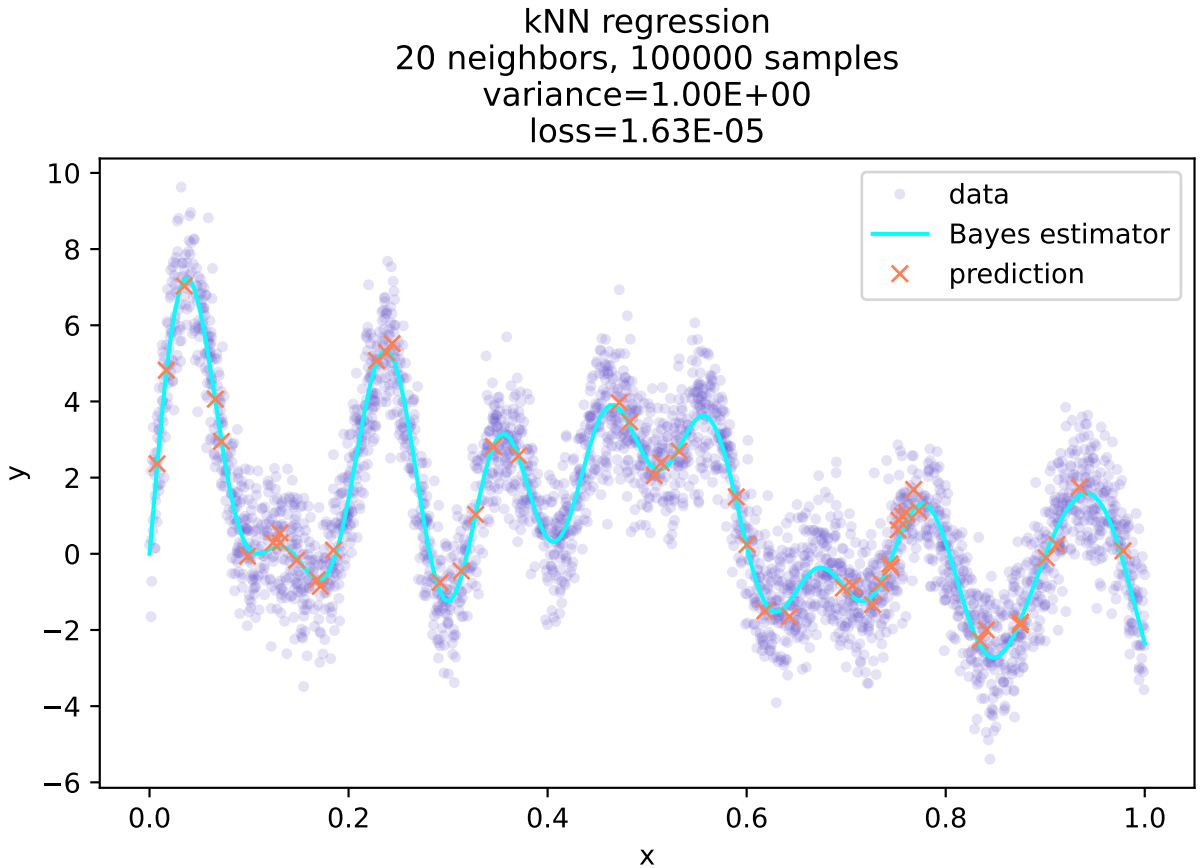
Machine Learning Tek 5



Textbooks

Textbooks are better than slides!

`https://github.com/nlehir/ML_tek5/blob/main/references.pdf`

FTML

I also make a course at Epita, you can find some material there and many practical sessions
`https://github.com/nlehir/FTML`

AI and the Turing test

- It is reasonable to say that the problem of AI was born at the same time as that of **Computer Science**. One of the founders of Computer Science is Alan Turing (1912-1954).
- In 1950, with the article **"Computing machinery and intelligence"** [Turing, 2009], Turing introduces the **Turing test**.
- One of the forms of a Turing Test is a game in which a computer tries to behave like a human by answering questions.

Turing, A.M. (1950). Computing machinery and intelligence. Mind, 59, 433-460.

COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. The Imitation Game

I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think." The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous, If the meaning of the words "machine" and "think" are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, "Can machines think?" is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the 'imitation game." It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in

AI and Turing test

- Importantly, Turing stresses that the terms "think", and "machines" are **ambiguous**.
- Many questions regarding our own "intelligence" and "consciousness" as humans remain open today.
- AI is a slightly ill-defined concept.

Moravec Paradox

- It is harder to emulate or simulate simple sensorimotor capacities than level abstract reasoning.
- ie : the sensorimotor system of an ant is harder for us to emulate than beating the world champion at chess (Deep Blue, 1997).

Conscious reasoning (1950's)

Broadly speaking, the first phases of research in AI were more dedicated to emulating the **conscious brain.**

- logic programming (Prolog)
- expert systems

History : AI Winter (1970's)

- In 1974 fundings for AI reseach started being cut by goverments because of unsuccessful projects.
- there were actually several so-called "AI winters".

Deep Blue (1997)

- Beat the world chess champion Gary Kasparov in 1997. This was an important milestone back then.
- 

Figure – Deep Blue

- Despite its name, Deep Blue is **not related** to Deep learning !

Decision trees and limitations

One of the underlying mechanisms of Deep Blue were **Decision Trees** / minimax algorithm.

- Let p be the **branching factor** of the tree. (Here, the average number of children at each node)
- Let d be the **depth** of the tree.
- What is the order of magnitude of the number of nodes in the tree ?

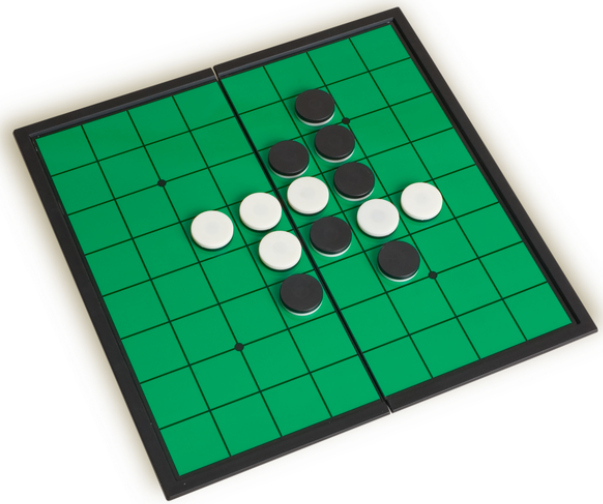
Limitations of the method

- Let p be the **branching factor** of the tree of actions. (Here, the average number of children at each node)
- Let d be the **depth** of the tree.
- What is the order of magnitude of the number of nodes in the tree ?
- So in order to run the minimax algorithm needs to perform $\mathcal{O}(p^d)$ evaluations.

Exercise

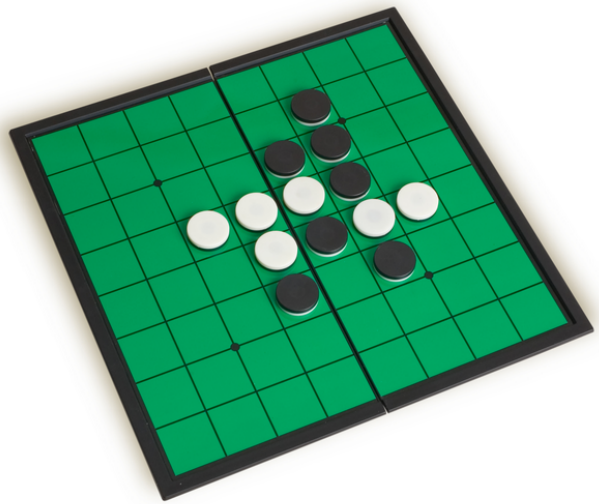
Exercice 1 : **Computation time**

- In the **Othello game**, the average number of actions in each state is around 8.
- We assume that the evaluation for one leaf node takes 1×10^{-6} seconds.
- How long would be the search of the minimax if we look 10 actions ahead ?



Exercise

- In the **Othello game**, the average number of actions in each state is around 8.
- This duration is too large for practical applications.



- In AlphaGo, the algorithm used was really different (Deep Reinforcement Learning.)

2000's and 2010's

- Around the 2000's, there has been a shift in the focus of AI.
- **Machine learning** and data science have become prominent research and industrial topics.
- We could argue that emulating the **unconscious brain** is more of a focus today.
- An important contemporary and future trend in research might be to connect research on emulating the conscious brain and research emulating the unconscious brain.

Interesting slides on AI history

https://home.mis.u-picardie.fr/~furst/IA.php

Machine learning : proposition of definitions

- (proposed definition of) **learning** : "Modification of a behavior, based on a life experiment"
- **Machine learning** : "artificial intelligence which can learn and model some phenomena without being explicitely programmed".
- In Machine learning, some parameters are learned in an **semi-automatic way**, based on a **dataset** in order to solve a problem or to optimize a solution.

Classical programming vs ML

- Classical program : predict the total amount of money spent, based on the number of fruits bought and the price of each individual fruit (a summation is enough)
- ML program : predict which fruit a person buys, based on a database of customers and on some information about this person (e.g. buying log, age, profession).

Use cases of ML

ML is useful for problems :

- that we cannot solve direcly thanks to an explicit representation (such as the amount of money spent in the previous first example).
- that we could solve explicitely, but the computationnal ressources would be too heavy (e.g. molecular dynamics), so we **approximate** a solution with ML.

Sometimes, ML solutions to a problem are effective, but the underlying mechanism is not perfectly understood (like in deep learning).

Machine learning

Machine learning (or "statistical learning") has received attention and funding since the beginning of the 2010's because it reached state-of-the-art efficiency on several problems such as :

- computer vision
- spam classification
- machine translation
- speech recognition
- self-driving cars
- (since the beginning of the 2020's) text / image / sound generation and prediction (sequence modelling)

Deep learning is involved in all these state-of-the-art solutions.

Example 1 : ImageNet

- A database of images (more than 15M), hand-annotated in order to indicate what objects are present in the image. More than 20000 categories of images.
- Contest : ImageNet Large Scale Visual Recognition Challenge.
- The best classification (top-5 error rate) went from 25% in 2011 to $\simeq$ 15.3% in 2012.
- The technology used was deep learning, exploiting GPUs (AlexNet).

`https://paperswithcode.com/sota/image-classification-on-imagenet`

Example 2 : AlphaGo

- In 2015 : beats a professionnal player. In 2017 : beats the world champion.
- Uses several technologies : among them **Deep reinforcement learning.**
- Improvements : AlphaGo Zero, trained without a database of played games. In 2017, AlphaZero beats AlphaGo Zero after 3 days of learning.

Example 3 : AlphaFold

- Goal : to predict the spatial configuration of proteins, from their DNA sequence.
- Achieves a breakthgough performance on the CASP challenge :
  - 2018 : more than 50% GDT (Global distance test), whereas it was $\leq 40\%$ before then.
  - 2020 : $92,4\%$ GDT. At a $\geq 90\%$ score, the method is considered competitive with experimental methods..
- `https://alphafold.ebi.ac.uk/`
- Also based on Deep learning.

Deep learning ⊂ Machine Learning ⊂ AI

- Machine learning is not only about deep learning.
- Some chapters of the course will be dedicated to neural networks and deep learning, as it is a really important contemporary topic. But we will not focus only on deep learning, but rather on more elementary / fundamental tools.
- Globally speaking, ML has a strong **experimental** component in industrial applications.

Why was there a ML revolution ?

- Technical progress in the computing and storage capacities (notably : application of **GPUs** to deep learning).
- Increase in amount of **available data**. According to IBM, 10^{18} bytes are created each day.
- Progress in algorithmic methods to analyze the data.

ML revolution II

- The speed of evolution has increased again with LLMs (first Chat GPT release : 2022)
- Spectrum of activities between maths and computer science : the jobs linked to AI and ML evolve. We could argue that the balance is more shifted towards data engineering / devops / mlops / computer science in general than 20 years ago.
- In this course, we will mention LLMs, but we will not go in depth on the topic (also : I am not an LLM expert).

Textbooks

Introduction to AI and ML

Learning paradigms
- Supervised learning
- Unsupervised learning
- Reinforcement learning
- Examples

What makes ML a hard problem ?
- Overfitting
- Curse of dimensionality
- Optimization

Supervised learning

Predict an output from an input.

- discrete output : classification
- continuous output : regression

We learn from a dataset D_n of n **labeled examples** ;

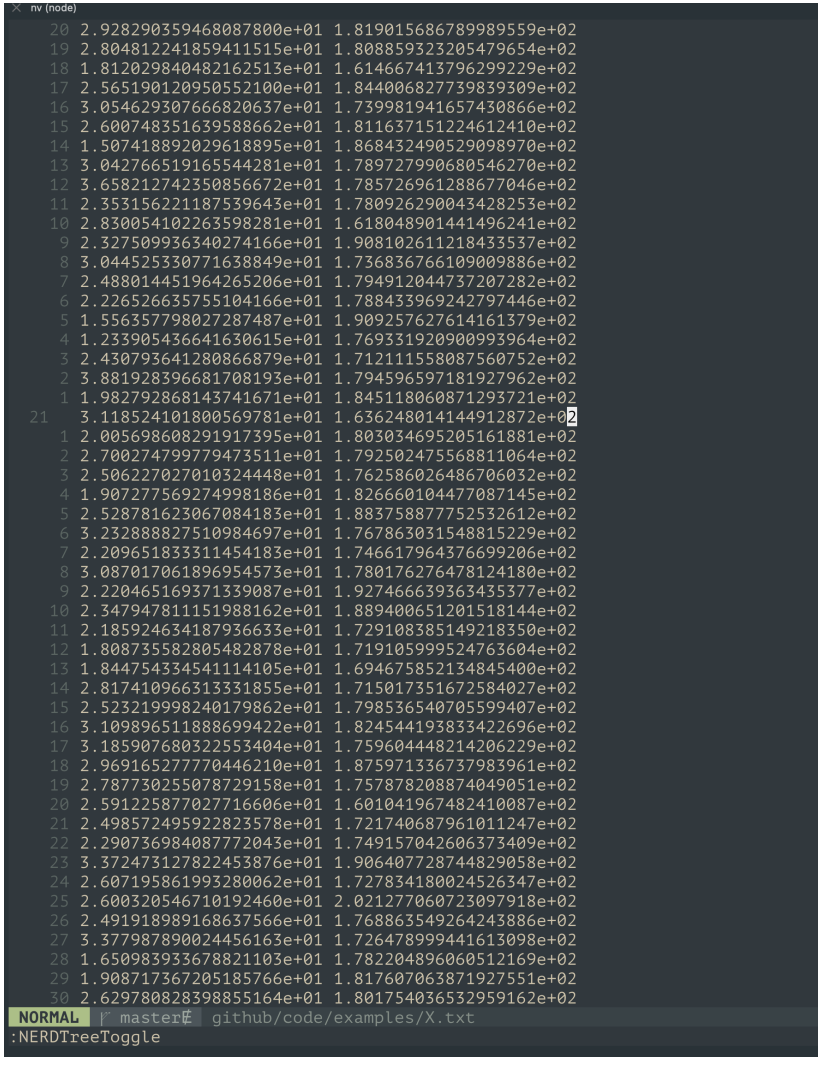
$$D_n = \{(x_i, y_i), i \in [1, n]\} \tag{1}$$

Please do not hesitate to ask for a clarification when a notation is unclear.

Supervised Learning : example of regression

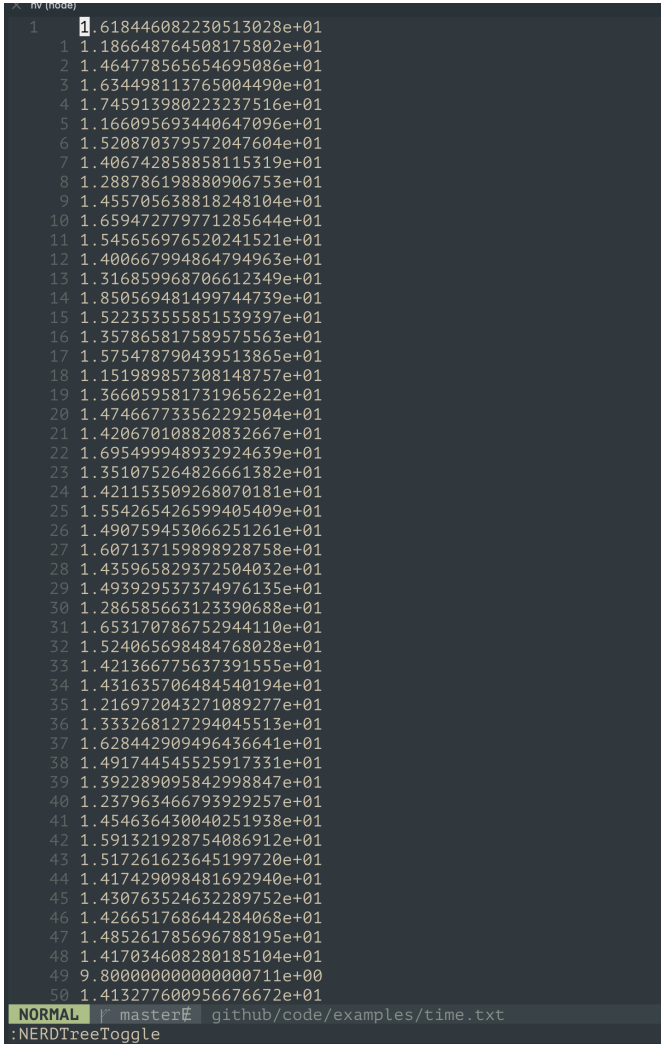
- x : age and height of a person. Here x, is a **vector** containing 2 **features**.
- y : record on a 100 meters track

Regression : raw inputs



Input data.

Regression : raw ouptuts



Exercice 2 :

Our objective is to build a function that predicts the output (100 meters track record) as a function of the **input** (height and age of the person).

What order of magnitude of precision do we expect to obtain with this example ?

Random variables

In a machine learning problem, there is (almost) always a statistical error associated with a prediction. Our goal is to minimize it ! Datasets should be seen as **sampled from a random variable**. Hence, **any quantity derived from the dataset** should be analyzed as a random variable as well.

Unsupervised Learning

- From a number of samples x_i, we want to retrieve information on their structure : **modelisation**.
- - density estimation
  - clustering
  - dimensionality reduction / computation of embeddings

Density estimation

- density estimation : anaylze the distribution of the data.

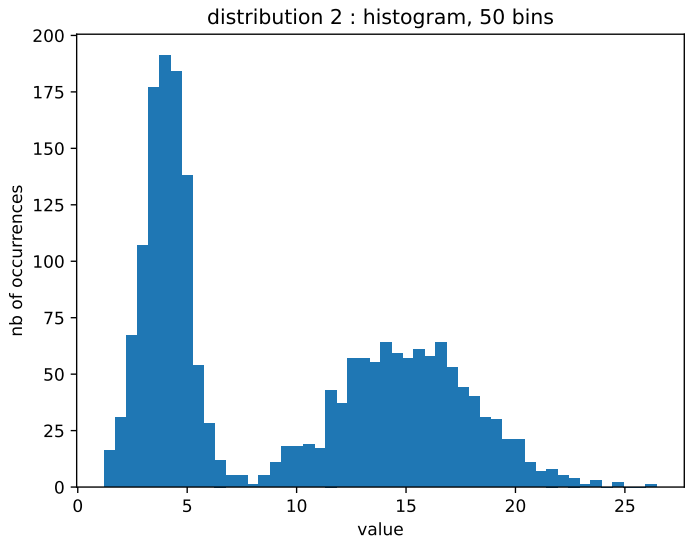


Figure – Histogram of random data

Clustering

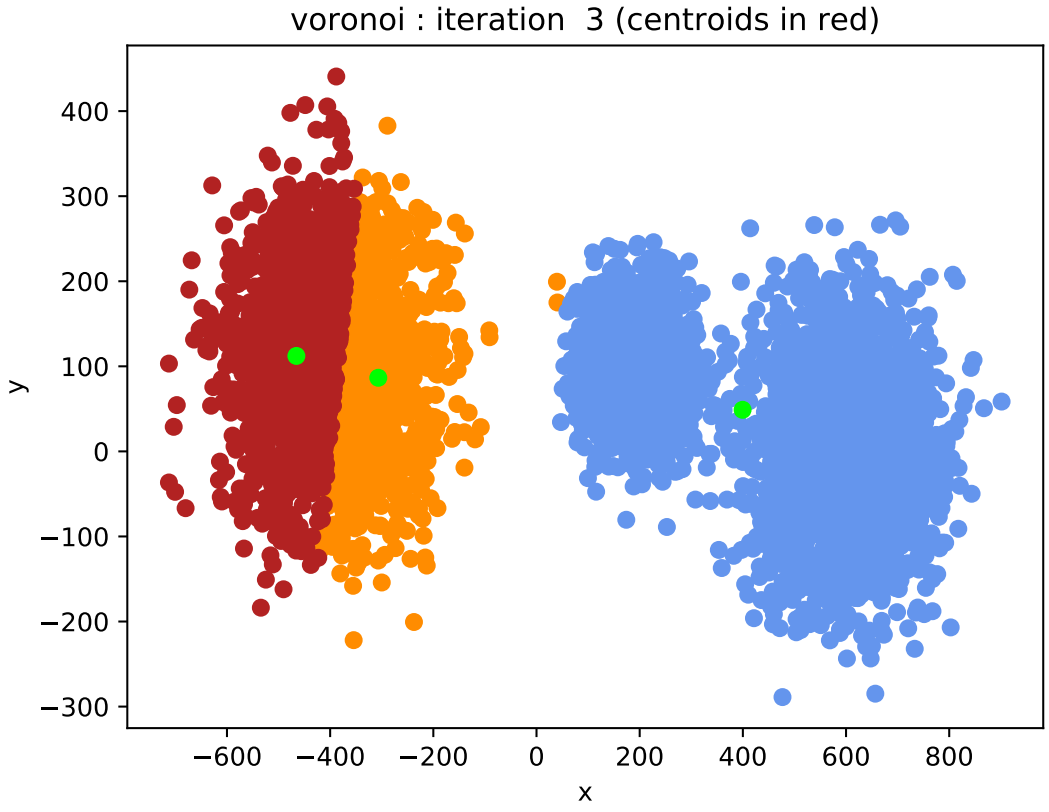


Figure – K-means clustering

Dimensionality reduction

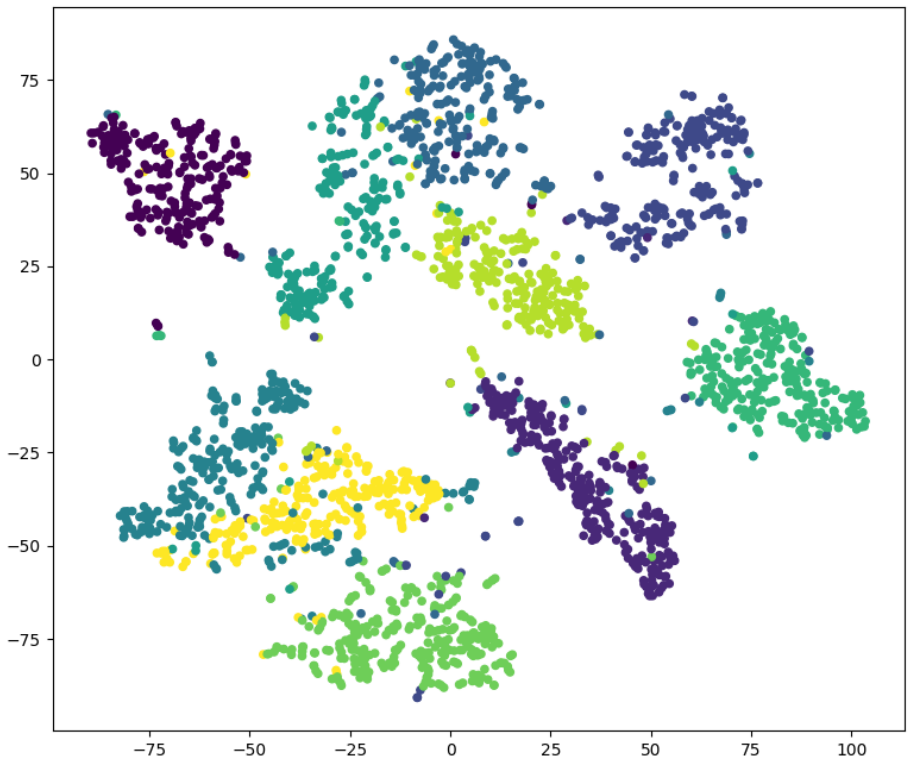


Figure – t-SNE projection of the MNIST dataset.

Reinforcement Learning

- A **more general paradigm** that describes an **agent** in a **world**.
- The standard formalization was the one proposed by Richard Sutton [Sutton and Barto, 2018].
- At each time, the world is in a state s. An agent perfoms an **action** a according to a **policy** π. When performing an action, the agent receives an **reward** r.

Reinforcement Learning applications

- Example : some Chessplayers, AlphaGo, automatic vacuum cleaner, self-driving cars (trajectory optimization, parking, etc.)
- healthcare : `https://arxiv.org/pdf/1908.08796.pdf`
- `https://deepmind.com/blog/article/alphago-zero-starting-scratch`

Reinforcement Learning

- At each time, the world is in a state s. An agent perfoms an **action** a according to a **policy** π. When performing an action, the agent receives an **reward** r.
- The agent wants to learn an **optimal policy**, meaning the policy that maximises its reward (or rather the **aggregation** of rewards).

This paradigm has many variants

- State s, action a, policy π, reward r.
- Is the policy **deterministic** ? Is it **stochastic** ?
- Does the agent have a **model** of its environment ?
- How many steps ahead should the agent look ?
- All these conditions change the way the problem should be addressed and solved.

- These paradigms can be mixed. For instance,
  - unsupervised learning can be used in a supervised learning problem (semi-supervised learning)
  - unsupervised learning and supervised learning can be used in a reinforcement learning problem
  - **self-supervised learning** creates labels from the dataset itsself.
- Given a problem and a dataset, formalizing a problem as a supervised learning or unsupervised learning task is a **choice**.
- Often, obtaining labels for all samples is a hard, long, or even impossible process : e.g. labelling millions of images. Importantly, the dataset collection, preparation or preprocessing is at least as important as the algorithms that are run on it !

Example I : what type of learning is it ?

Predict the winning team of an NBA game at half-time.

- Dataset : 15 years of games (comments, text) : approximately 17000 games.
- The dataset is preprocessed to have as an input a time-series : each time contains the score **and** 10 technical features (rebounds, etc.). So for each time the dimension is 11. Each game is a matrix of size 1440×11, reorganized as a line vector.
- Output : Receiving team wins or looses
- Evaluation metric : "0-1" loss (1 if error, 0 otherwise)

Example II : what type of learning is it ?

Predict the quantity of oil in a rock.

- Input : tomographic image of a rock.
- Output : material of the rock, average presence of residual oil in the rock.

Example III : what type of learning is it ?

- Input : database of clients of an insurance comrany
- Output : groups of clients, based on customer behavior, in order to propose targetted products for the identified groups.

Example IV : what type of learning is it ?

Detect issues in wind farms.

- Input : sensors on the wind turbine (wind direction, air temperature, electric tension, rotation speed, component temperature, etc.) as a time series. Each step represents 10 minutes (several years).
- Output : Power generated by the turbine.
- Evaluation metric : MAE (mean absolute error).

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What makes ML a hard problem ?
- Overfitting
- Curse of dimensionality
- Optimization

Overfitting

If the information that as been learned during the learning phase does not generalize (or is not as relevant) to new, unseen samples, this is called **overfitting** (more details in the next chapter).
Remark : if there was no noise, or apparent randomness in the data, there would be no overfitting !

Curse of dimensionality

Two numbers are important in machine learning :

- n : number of samples
- d : dimension (number of features) of a unique sample

Both can be large and prohibitive for some algorithms. Typically

- the larger they are, the slower algorithms are
- the smaller n is, the higher the risk of overfitting is.
- the higher d is, the higher the risk of overfitting is.
- the higher d is, the harder is is to learn.

Curse of dimensionality

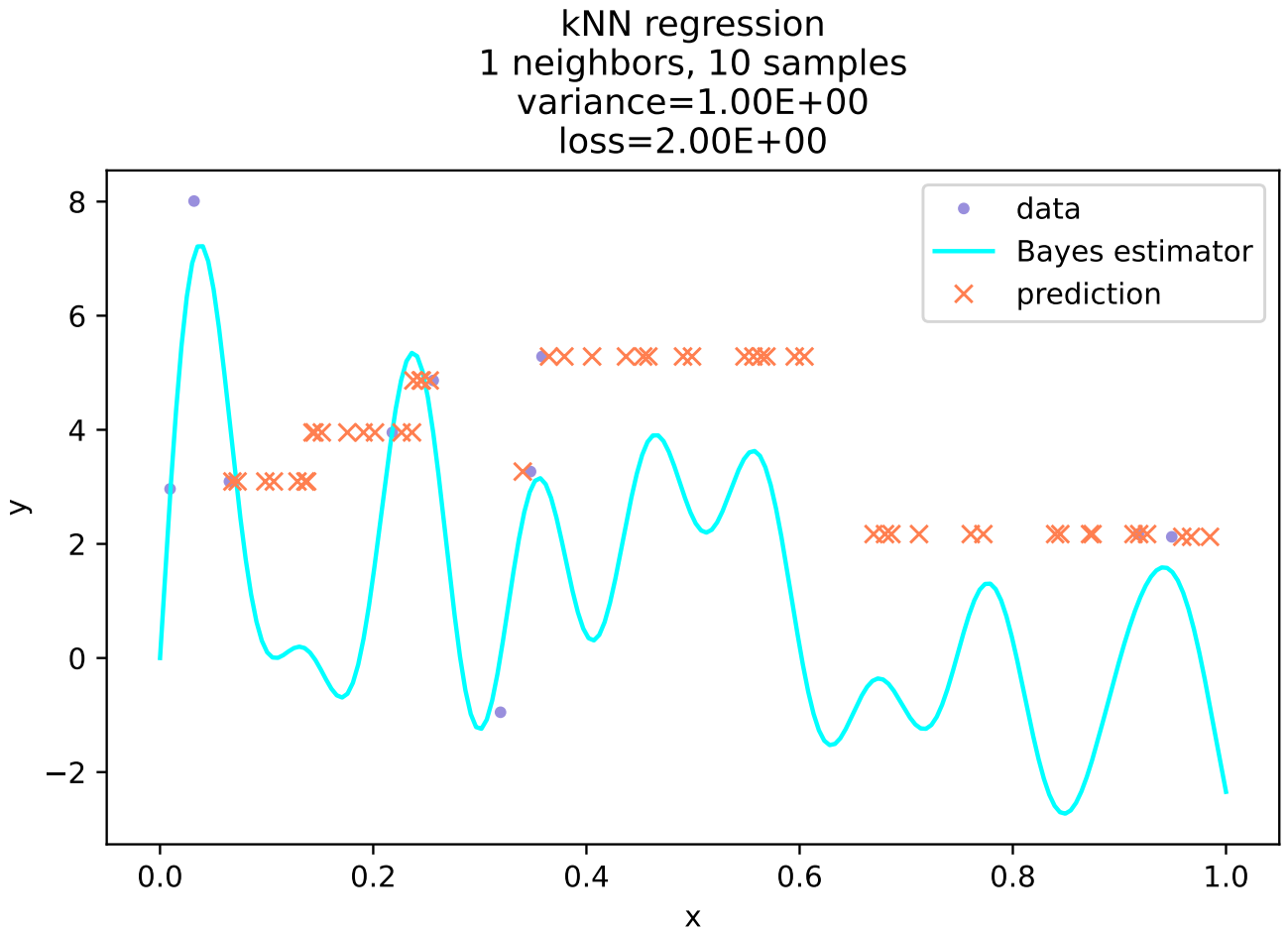
Many datasets have large n and / or d.

- images
- DNA sequence
- text
- audio/video

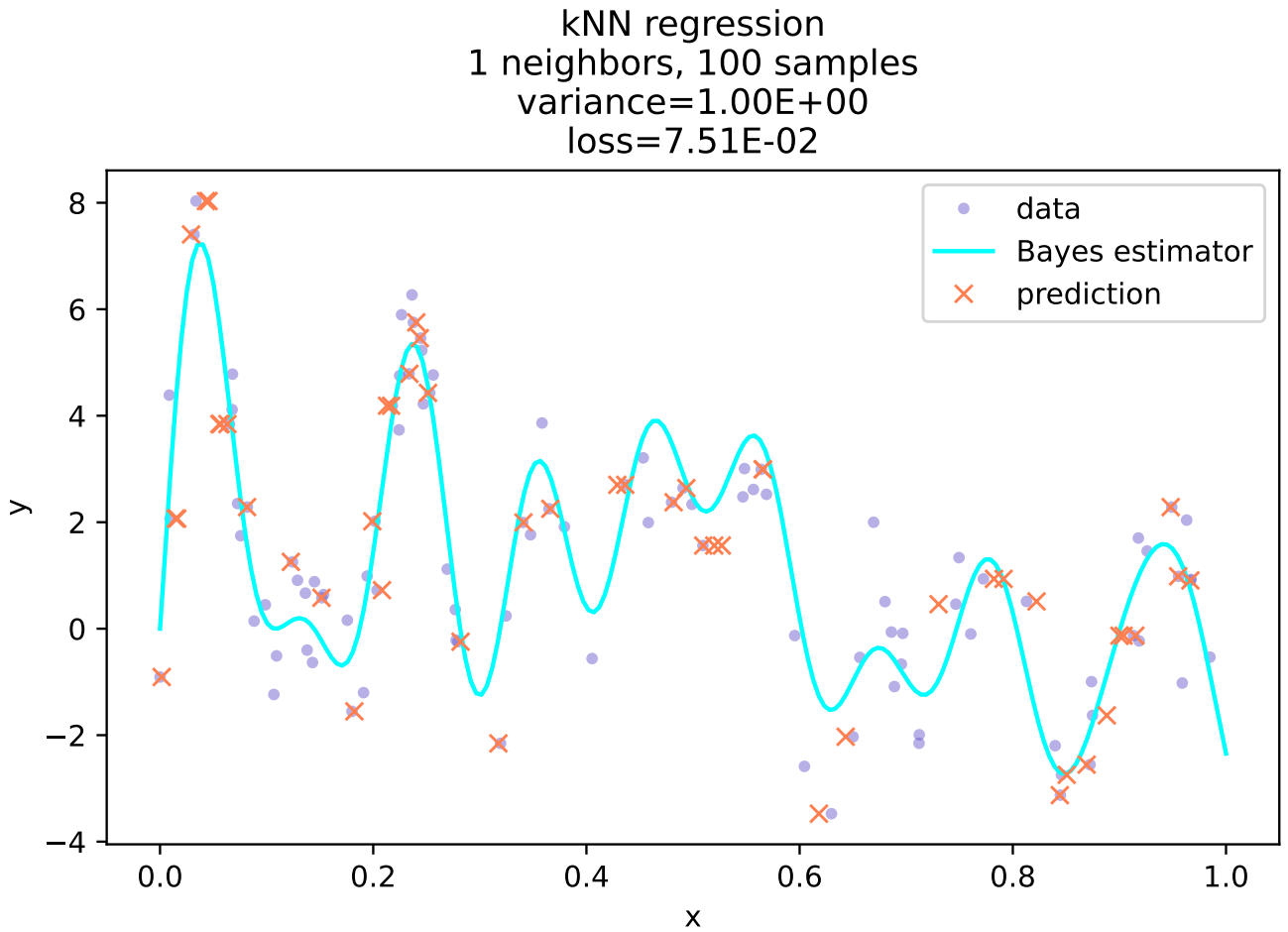
Local averaging

The curse of dimensionality is the reason why **local averaging** is most often inefficient then d is large.

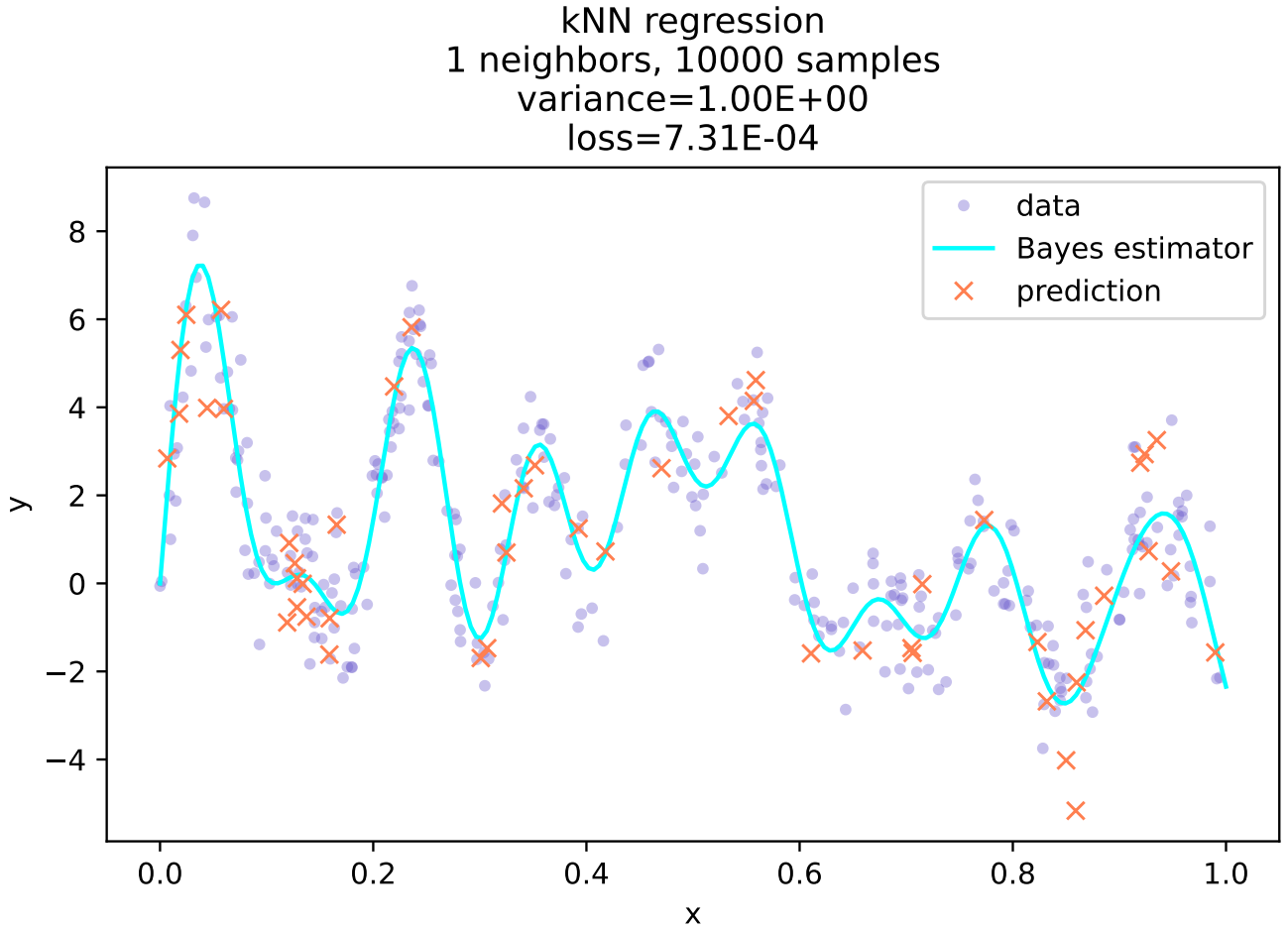
Local averaging



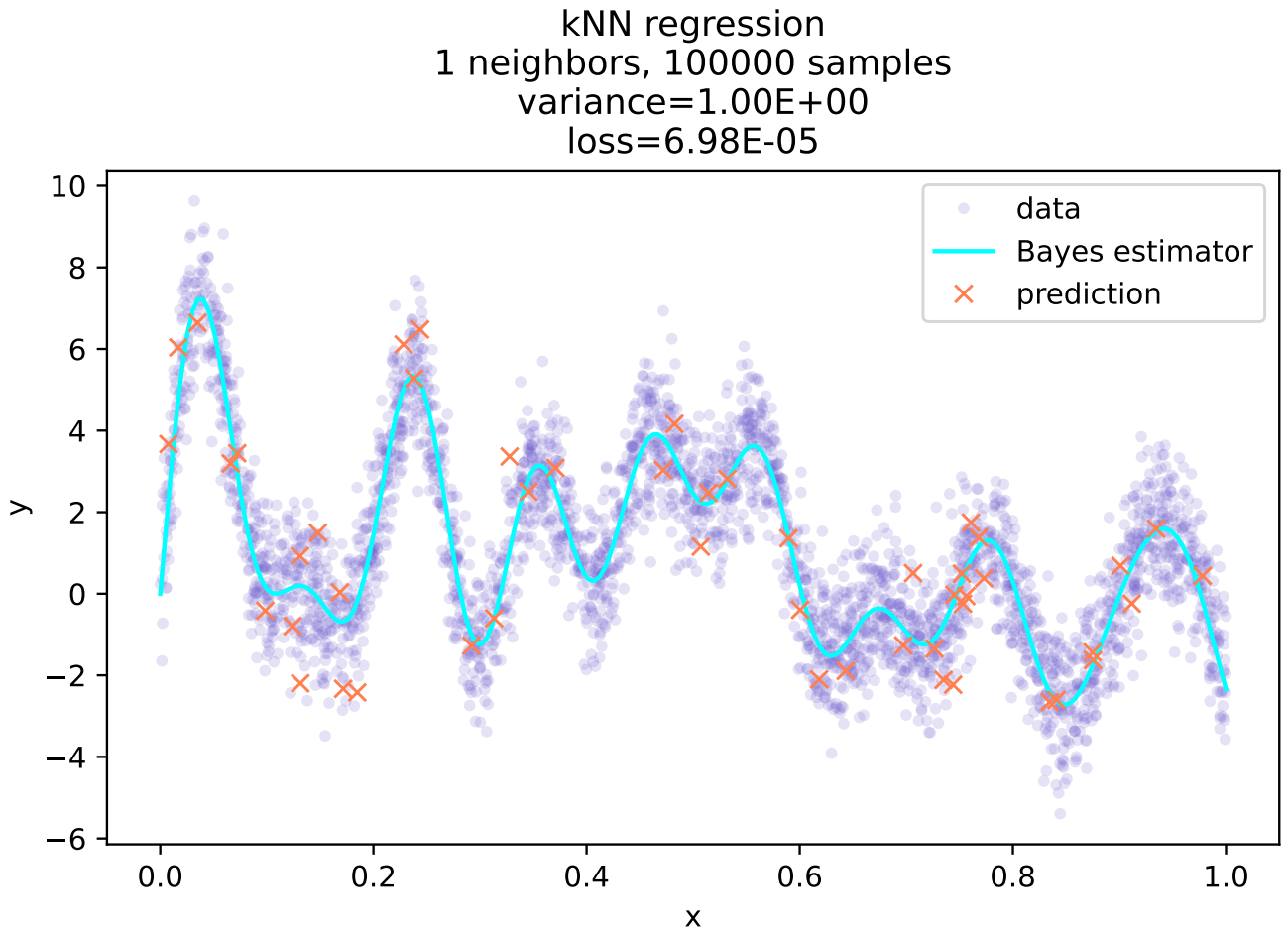
Local averaging



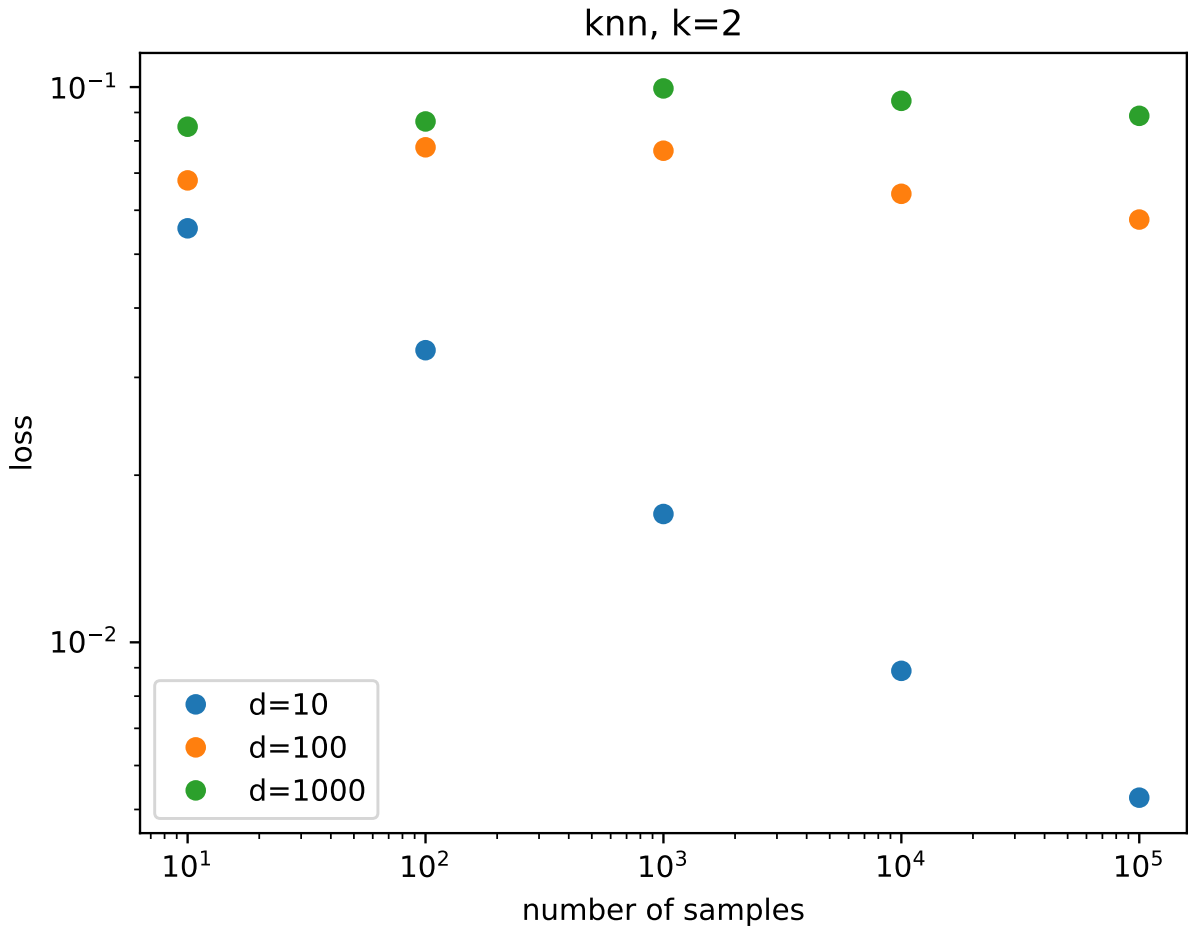
Local averaging



Local averaging



In d dimensions

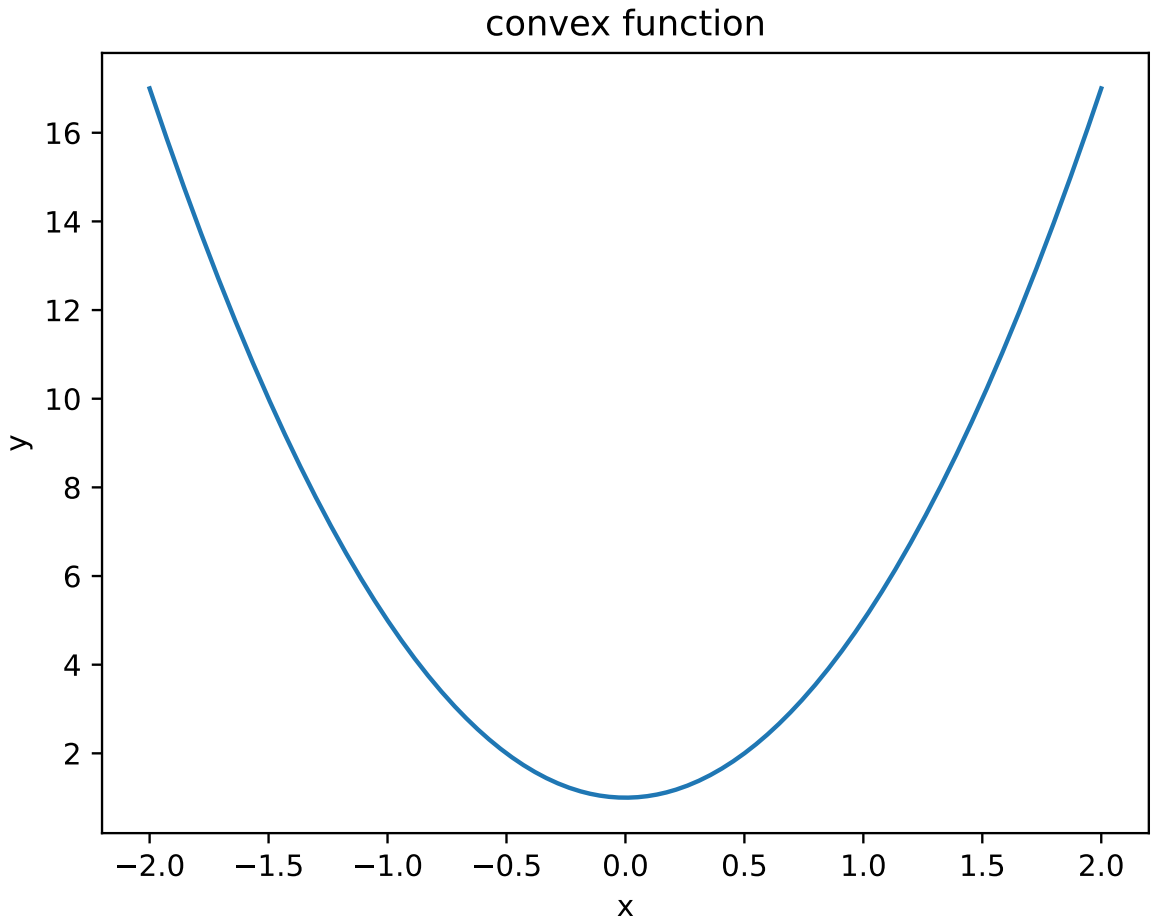


Optimization

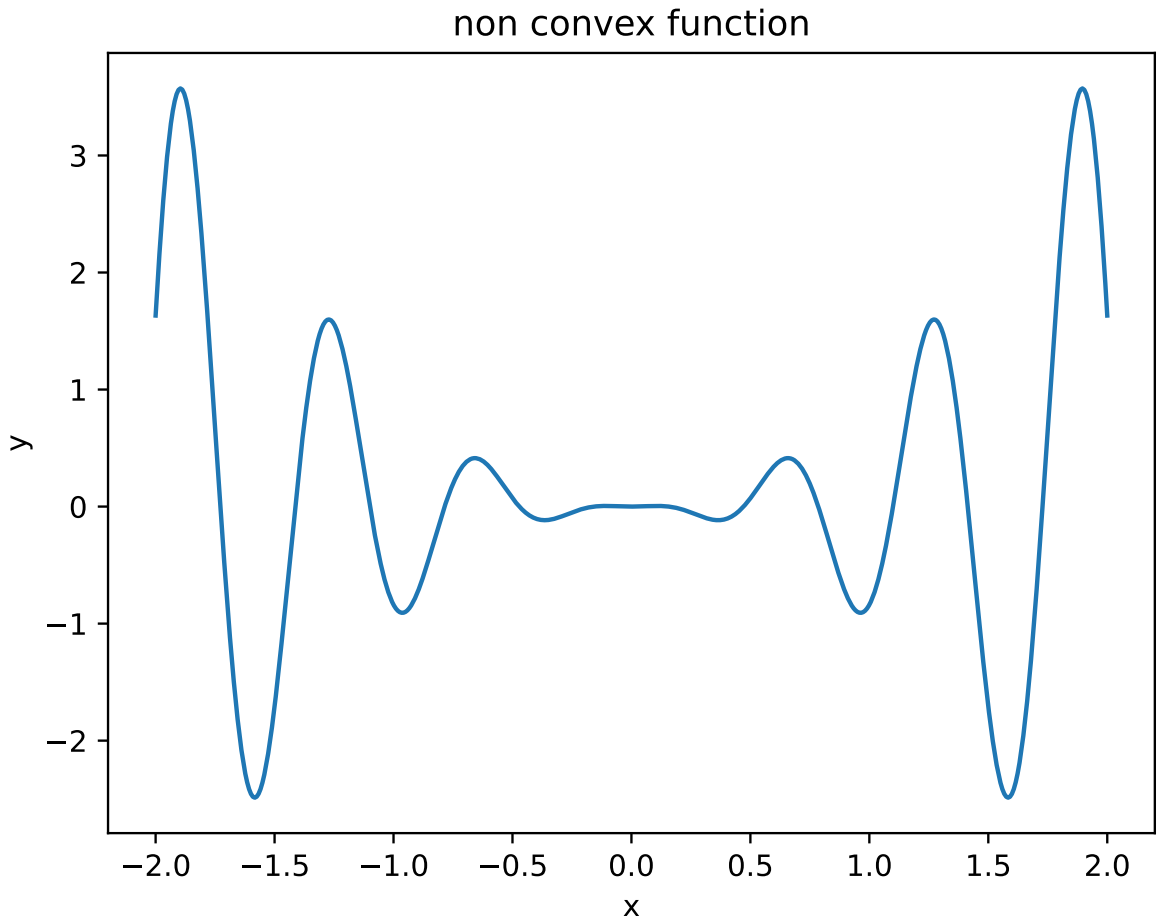
As we will discuss shortly, learning is reformulated as an **optimization problem**. We often have to optimize functions that are **hard** to optimize.

- non convexity
- non linearity
- high dimension
- instability of estimators

Convex function



Non-convex function



The state of ML

ML is an active research area. One of the biggest effort is trying to understand the performance of deep learning.

- they are non convex
- but we experimentally manage to optimize them so that they have a state-of-the-art generalization error

In the case of neural networks with 1 hidden layer, the convergence of gradient descent to a global minimizer of the objective function has been proven in 2018 in a specific case (large number of hidden neurons), using tools from optimal transport theory ([Chizat and Bach, 2018]).

The double descent phenomenon (we will discuss it again later)

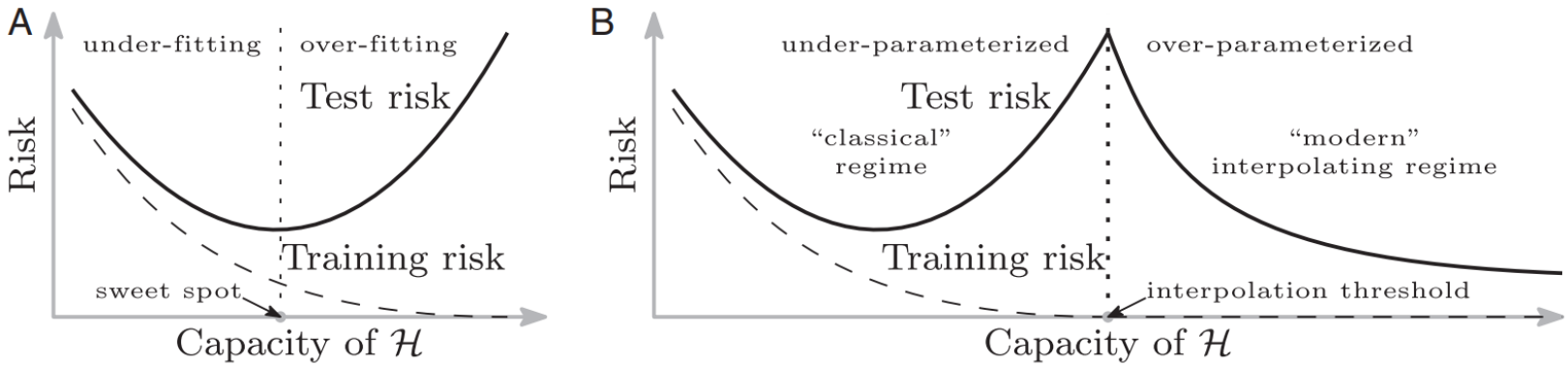


Figure – [Belkin et al, 2019]

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